

● RESEARCH GRADE · 99%+ PURITY · USA-BASED

RESEARCH REFERENCE & SAFETY DOCUMENTATION

5-Amino-1MQ



5-Amino-1-methylquinolinium · NNMT inhibitor · Research-grade reference material

CLASS

Small molecule

SCAFFOLD

Quinolinium salt

CATION FORMULA

$C_{10}H_{11}N_2^+$

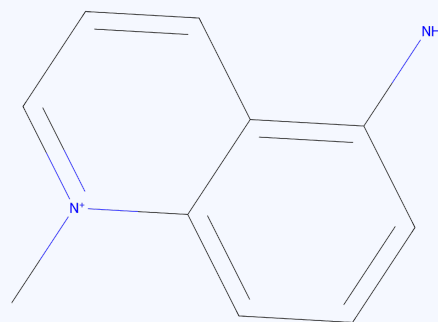
CATION MW

159.2 g/mol

Identity & Composition

5-Amino-1MQ is a synthetic small molecule — a quaternary ammonium salt built on the quinolinium scaffold — developed as a selective inhibitor of nicotinamide N-methyltransferase (NNMT). **It is not a peptide:** it has no amino-acid backbone, and its chemistry, mechanism and evidence base are those of a small-molecule enzyme inhibitor.

Product name	5-Amino-1MQ
Chemical name	5-Amino-1-methylquinolinium
Class	Small molecule; quaternary ammonium (quinolinium) salt
Target	Nicotinamide N-methyltransferase (NNMT)
Cation formula	C ₁₀ H ₁₁ N ₂ ⁺
Cation MW	159.21 g/mol
Salt form	Commonly supplied as the iodide salt — confirm counter-ion and salt MW against lot documentation
Appearance	Crystalline solid
Identified use	In-vitro research & development only



Molecular structure. 5-Amino-1-methylquinolinium cation (C₁₀H₁₁N₂⁺, 159.21 g/mol). The permanent positive charge on the quaternized ring nitrogen is intrinsic to the scaffold; the material is supplied as a salt with an accompanying anion.

Quantitation note. Because the compound is supplied as a salt, the cation mass (159.21) and the salt mass differ. Confirm which the certificate of analysis reports before preparing molar solutions — this is a common source of concentration error with quaternary ammonium reagents.

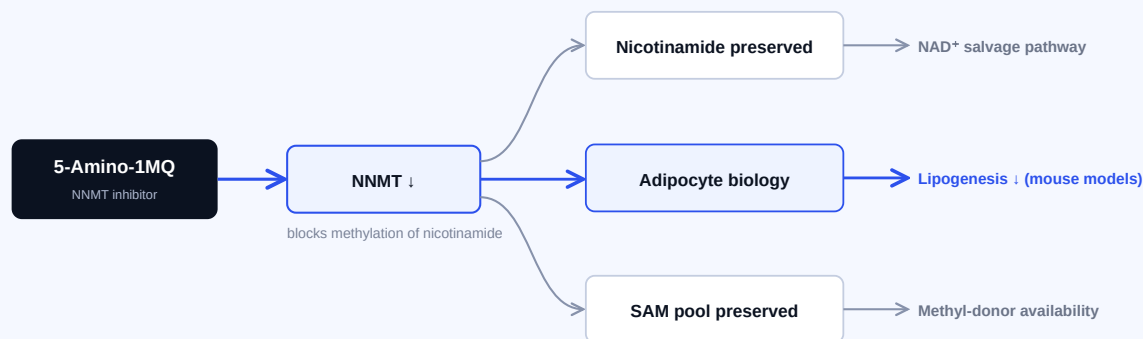
Research Background

NNMT catalyses the methylation of nicotinamide using S-adenosylmethionine (SAM) as methyl donor, producing 1-methylnicotinamide and SAH. It therefore sits at the intersection of two pathways: it consumes nicotinamide (a NAD⁺ precursor) and it consumes SAM (the cell's principal methyl donor). Inhibiting it is a way to probe both simultaneously.

Mechanism and preclinical findings

All findings below derive from cell-culture and rodent models. There are no human data.

- **NNMT inhibition.** Blocks the methylation drain on nicotinamide and SAM, with reported downstream effects on NAD⁺ salvage and methyl-donor availability.
- **Adipocyte models.** In 3T3-L1 preadipocyte culture, reported to inhibit differentiation and reduce lipid accumulation across a micromolar concentration range.
- **Diet-induced obesity models.** Neelakantan et al. (2019) reported reduced weight gain, decreased fat mass and smaller adipocytes in high-fat-diet mice **without differences in food intake** — mechanistically distinct from appetite-mediated compounds.
- **Metabolic parameters.** Improved glucose tolerance and insulin sensitivity reported in obese rodent models.
- **Tissue context.** NNMT is highly expressed in white adipose tissue and liver, and elevated expression has been observed in obese adipose tissue — the rationale for adipose as the primary research target.



Proposed mechanism investigated in preclinical (in-vitro / rodent) models. Schematic only; not a claim of established human effect.

What the Literature Supports — and What It Doesn't

The position here is simple and should not be overstated: 5-Amino-1MQ is a preclinical research tool. Its mechanism is coherent and its rodent data are internally consistent, but the human column of the evidence table is empty.

Evidence at a glance

IN-VITRO

Characterized

RODENT MODELS

Consistent

HUMAN TRIALS

None published

REGULATORY

Not approved

- **No published human clinical trials exist.** Every reported effect on body composition, glucose tolerance and adipocyte biology comes from mouse or cell studies.
- **Metabolic targets have a poor translation record.** Multiple prior anti-obesity mechanisms succeeded in rodents and failed in human trials; rodent adipose findings are weak predictors of human outcomes.
- The compound belongs to a **traceable medicinal-chemistry lineage** of methylquinolinium NNMT inhibitors, which is a genuine strength for research provenance.
- Reported quantitative effects vary substantially between secondary sources; **rely on the primary publications** rather than aggregated figures when designing experiments.

Appropriate characterization: a selective small-molecule NNMT inhibitor useful as a research tool for probing methylation flux, NAD⁺ salvage and adipocyte metabolism — with no human evidence and no established translational relevance.

Regulatory & status notes

US FDA	Not approved for any use. Not a dietary ingredient; not an approved drug.
Supply basis	Research reagent for in-vitro laboratory use only; not for human or veterinary administration.

Selected references

1. Kraus D, et al. Nicotinamide N-methyltransferase knockdown protects against diet-induced obesity. *Nature*. 2014. (foundational NNMT target validation)
2. Neelakantan H, et al. Selective and membrane-permeable small molecule inhibitors of nicotinamide N-methyltransferase reverse high fat diet-induced obesity in mice. *Biochem Pharmacol / Cell Chem Biol*. 2019.
3. Pharmacokinetics and tissue distribution characterization of 5A1MQ in mice. 2024.

Note. This Safety Data Sheet is compiled from public sources and weight-of-evidence assessment. Independent review by a qualified chemical-safety professional is recommended prior to use. Supplier identification and emergency-contact details must be completed by the supplier.

1 Identification

Product	5-Amino-1MQ (5-amino-1-methylquinolinium)
Class	Small molecule; quaternary ammonium salt (not a peptide)
Cation formula / MW	C10H11N2+ / 159.21
Salt form	Confirm counter-ion against lot documentation
Use	In-vitro research only
Restriction	Not for human/veterinary, food, drug, cosmetic, clinical, or therapeutic use
Supplier	
Emergency	

2 Hazard Identification

Not classified based on currently available data; toxicological data are limited and absence of classification is not a determination of absence of hazard. Quaternary ammonium compounds as a chemical class can be irritating to skin, eyes and the respiratory tract — a qualified reviewer should confirm whether any class-based classification applies. **Precautionary:** P261, P264, P280, P302+P352, P305+P351+P338, P501.

3 Composition

Single-substance product, >98% research grade, supplied as a salt. Residual synthesis reagents and solvents may be present. See lot COA for impurity profile and salt form.

4 First-Aid Measures

Eyes: rinse cautiously with water for several minutes; remove contact lenses if present and easy to do. **Skin:** wash with plenty of soap and water; remove contaminated clothing. **Inhalation:** move to fresh air. **Ingestion:** rinse mouth; do not induce vomiting unless directed. Seek medical advice if irritation persists.

ANVIL COMPOUNDS

5 Fire-Fighting

Media: CO₂, dry chemical, foam, water spray. Combustion may produce CO, CO₂, nitrogen oxides, and hydrogen halides if a halide salt. Use SCBA and full protective gear.

6 Accidental Release

Avoid dust formation; ventilate; use PPE per §8. Prevent entry to drains/waterways — quaternary ammonium compounds can be harmful to aquatic organisms. Collect in a closed labeled container for disposal per §13.

7 Handling & Storage

Handle as a biologically active small molecule of incompletely characterized toxicity; avoid dust generation and skin contact. Weigh in a fume hood or ventilated enclosure. Store tightly closed, dry, protected from light; refer to lot documentation for recommended temperature. Observe COA retest date.

8 Exposure Controls / PPE

No established OEL; control to ALARA. Local exhaust or fume hood for the dry solid. Nitrile gloves; ANSI Z87.1 eye protection (goggles where splash possible); lab coat; respiratory protection where dust control is insufficient. Eyewash/safety shower per ANSI Z358.1.

9 Physical & Chemical Properties

State	Crystalline solid
Appearance	Off-white to yellow
Odor	Odorless
Solubility	Water soluble (salt)
Cation MW	159.21 g/mol
MP/BP/Flash	Not determined

10 Stability & Reactivity

Stable under recommended storage conditions. Avoid heat, moisture, direct sunlight, open flame. Incompatible with strong oxidizers and strong bases. Decomposition: CO, CO₂, NO_x, and hydrogen halides if a halide salt. No hazardous polymerization.

11 Toxicological Information

Not fully characterized. No authoritative LD₅₀/LC₅₀ identified for this substance. Skin/eye: treat as potentially irritating (quaternary ammonium class). Sensitization: cannot be ruled out. Mutagenicity, carcinogenicity, reproductive toxicity, STOT, aspiration: no authoritative substance-specific data; not classified based on available data; hazards cannot be excluded. Avoid exposure if pregnant, nursing, or planning pregnancy.

12 Ecological Information

No substance-specific ecotoxicity data identified. Quaternary ammonium compounds can be toxic to aquatic organisms; as a precaution avoid release to surface water, soil, drains or sewers. A qualified reviewer should confirm whether aquatic-toxicity classification applies.

13 Disposal

Dispose per applicable local/state/federal regulations; US per 40 CFR 261–270 (RCRA). Do not dispose into sewers/waterways. Use a licensed waste disposal company.

14 Transport Information

Not regulated as dangerous goods under DOT (49 CFR), IATA or IMDG based on current classification. UN number: not applicable. Shipper must independently verify.

15 Regulatory Information

US TSCA: may qualify for R&D exemption (40 CFR 720.36). OSHA HazCom 2012 aligned with GHS Rev. 8. EU REACH: supplied for SR&D use; verify exemption. Not a legal determination of regulatory status.

16 Other Information

Revision [DATE] · Version 1.0. Independent professional review recommended prior to use — note the quaternary-ammonium class-classification item above. Sources: PubChem; peer-reviewed literature; OSHA HazCom 2012; GHS Rev. 8. Provided without warranty. For scientific R&D use only.

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